Project: Telco Customer Churn - Exploratory Data Analysis and Data Preprocessing

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Course: ITAI 1371

Dataset Proposal

Dataset Description

For this project, our group selected the Telco Customer Churn dataset, which contains information about customers of a telecommunications company and whether they stayed with the company or left. The original dataset contains 7,043 rows, with each row representing a customer. It includes numerical features such as tenure, MonthlyCharges, and TotalCharges, as well as categorical features such as Contract, InternetService, and PaymentMethod. The target variable, Churn, indicates whether a customer left the company.

Before using the dataset for machine learning, we need to inspect its structure and data quality. Our preparation work includes checking for missing values, blank entries, and duplicate records, converting data into appropriate formats, and preparing numerical and categorical features. We also use visualizations to understand customer information and identify patterns in the data.

Project Proposal

The main goal of our project is to prepare the Telco Customer Churn dataset for a machine learning classification model that can predict whether a customer is likely to leave the company. By studying customer information such as contract type, tenure, and monthly charges, we hope to identify patterns that may be related to customer churn.

Our planned workflow includes cleaning the data, encoding categorical variables, scaling numerical features when needed, and separating the data into training and testing sets using a 70/30 split. We will also examine the distribution of churn and non-churn customers. If class balancing is needed, we plan to use RandomOverSampler on the training data only, while keeping the testing data's original distribution for evaluation.

Through this project, we aim to build a well-prepared dataset for future model training and evaluation. We also want to understand how data cleaning, visualization, feature preparation, class imbalance, and preventing data leakage can affect the reliability of machine learning results. This project will help our group gain practical experience preparing real-world data for a customer churn prediction problem.